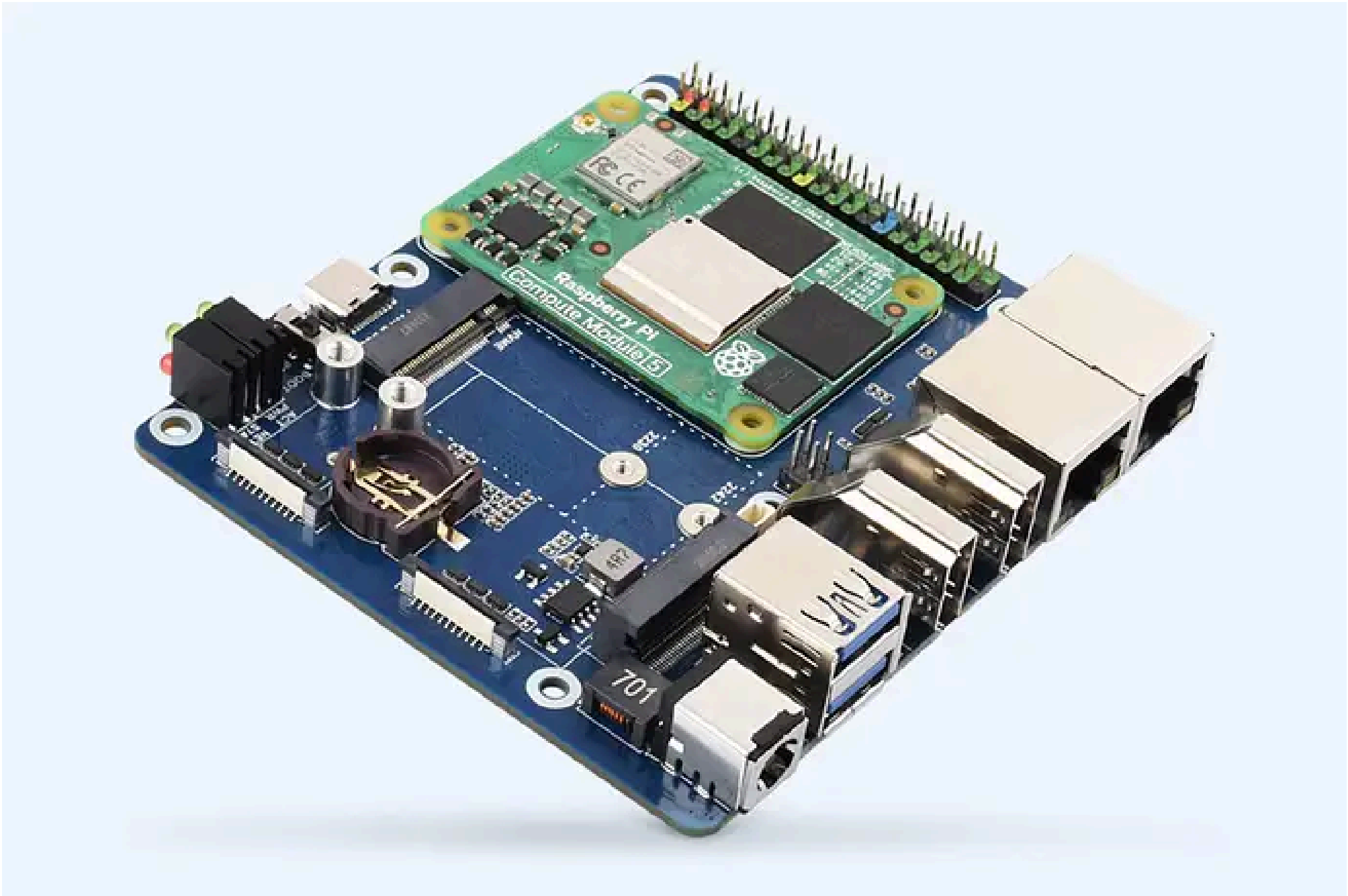
Waveshare CM5-DUAL-ETH-4G/5G-BASE is an industrial carrier board designed specifically for the Raspberry Pi Compute Module 5 (CM5). It serves applications such as industrial automation, edge computing, and IoT gateways, providing reliable connectivity and robust hardware integration.

Compared with the **previous CM5-DUAL-ETH-BASE (Ethernet-only)**, this model adds a 4G LTE / 5G cellular expansion interface while retaining dual Ethernet—offering flexible networking for environments where wired connectivity is not always available.

The board also integrates:

- Metal enclosure (optional BOX-A version) for mechanical protection
- Active cooling fan (optional) for thermal stability under continuous load
- Wide power input: **7–36V DC,** suitable for diverse industrial environments
- Rear I/O port arrangement for cleaner wiring and easier cabinet integration



Key Features and Functional Design

1. Network Connectivity: Multi-Scenario Communication

Dual Ethernet Architecture

- **1 × Gigabit RJ45 (10/100/1000 Mbps)**
- **1 × 2.5Gbps RJ45 (10/100/1000/2500 Mbps)**

The dual-speed configuration covers both standard industrial networking and bandwidth-intensive applications such as:

- High-resolution IP camera streaming
- Large-scale data acquisition
- Edge computing workloads requiring fast upstream bandwidth

PoE capability (via add-on module, 802.3af) allows both network + power through one cable, eliminating additional power wiring.

Cellular Connectivity: 4G/5G Ready

Using the M.2 B-Key slot, the board supports 4G/5G modules such as:

- SIM7600G
- SIM8200EA-M2
- SIM8262E-M2

Combined with the onboard **Nano-SIM** slot, this enables:

- Mobile device deployments (vehicles, remote sites)
- Wired + wireless failover redundancy
- Outdoor IoT terminal connectivity without infrastructure dependency

If wired Ethernet is unavailable, cellular automatically ensures data continuity.

2. Multimedia, Storage, and Expansion

Display and Camera Interfaces

- **2 × HDMI 2.0,** supporting 4K@60Hz output
- **2 × MIPI DSI / CSI** multiplexed ports (for display or camera)

This makes the board suitable for:

- Industrial HMI panels
- Vision AI (inspection, recognition, tracking)
- Multi-screen dashboards and data visualization

Storage and External Expansion

- M.2 M-Key (PCIe Gen2/Gen3 x1)
  - NVMe SSD support
  - Optional AI accelerator support
- MicroSD slot (for CM5 Lite only)
- 2 × USB 3.2 Gen1 Type-A (5Gbps)
- 40-pin GPIO (I²C, UART, SPI, etc.) for sensors and actuators

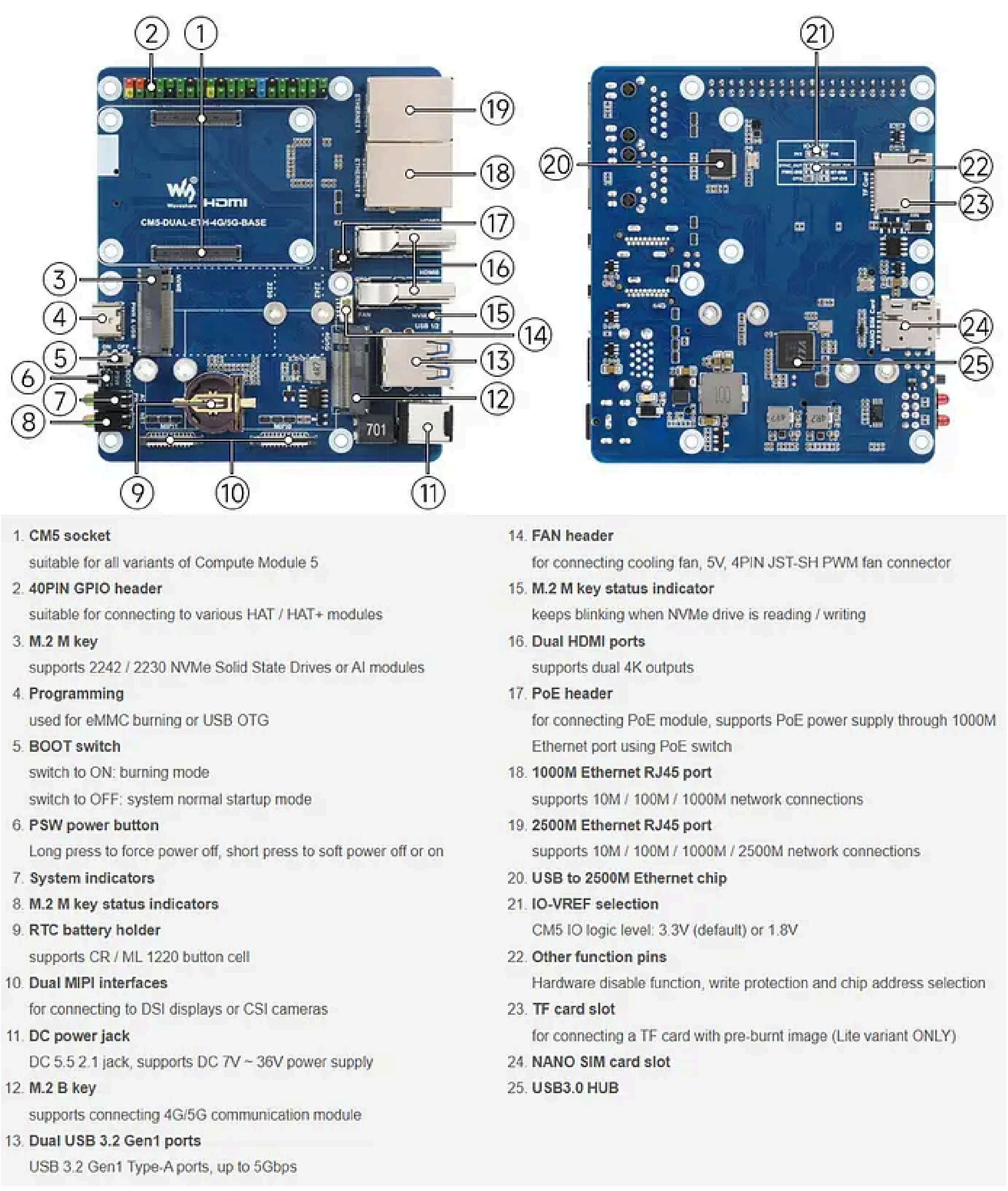
3. Industrial-Grade Hardware Design

- Power input: 7–36V DC, USB-C 5V/5A, or PoE (add-on module required)
- RTC battery backup (CR/ML1220)
- PWM controllable fan port
- EEPROM + hardware write-protection
- Configurable logic level: 3.3V / 1.8V

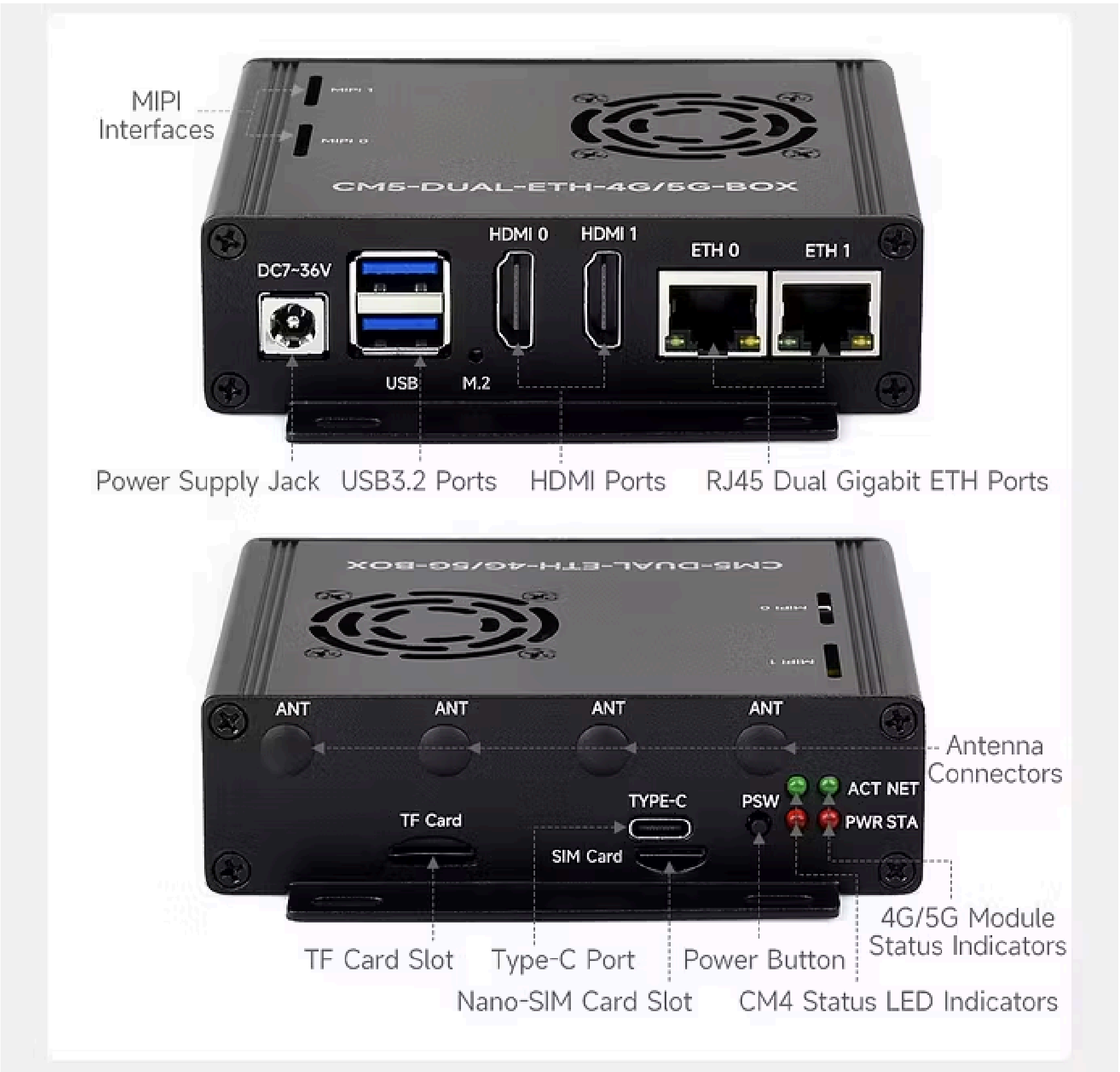
Optimized for industrial cabinets, long-term 24/7 operation, and vibration-prone environments.

Hardware Specifications

Category	Details
Compatibility	Raspberry Pi Compute Module 5 (CM5)
Storage	M.2 M-Key: NVMe SSD or AI accelerator module (PCIe Gen2/Gen3 x1) MicroSD slot: CM5 Lite only
Display	2 × HDMI 2.0, up to 4K@60Hz 2 × MIPI DSI (shared with CSI)
Camera	2 × MIPI CSI (shared with DSI)
Audio	HDMI audio output
USB	2 × USB 3.2 Gen1 Type-A (5Gbps) 1 × USB 2.0 Type-C (flashing / power input)
Networking	1 × Gigabit Ethernet RJ45 (10/100/1000 Mbps), PoE (802.3af) via external module 1 × 2.5GbE RJ45 (10/100/1000/2500 Mbps) Wireless: CM5 onboard Wi-Fi 5 / Bluetooth 5.0 (optional) Cellular: M.2 B-Key (4G/5G module), Nano-SIM slot
Expansion	M.2 M-Key: NVMe SSD / AI accelerator M.2 B-Key: 4G/5G cellular module 40-pin GPIO (I²C/UART/SPI etc.)
Power	7V–36V DC input (5.5×2.1mm jack) USB-C 5V/5A input PoE power (via Gigabit port, add-on module required)
Dimensions & Mounting	Board: 99.6 × 87.0mm BOX-A enclosure: 104.8 × 90.0 × 29.7mm Wall/ DIN-rail mounting (BOX-A version)
Other Features	RTC battery holder (CR/ML1220) PWM fan control Boot mode switch 4 × status LEDs EEPROM + write protection 3.3V / 1.8V logic level switch



1. **CM5 socket**
   suitable for all variants of Compute Module 5
2. **40PIN GPIO header**
   suitable for connecting to various HAT / HAT+ modules
3. **M.2 M key**
   supports 2242 / 2230 NVMe Solid State Drives or AI modules
4. **Programming**
   used for eMMC burning or USB OTG
5. **BOOT switch**
   switch to ON: burning mode
   switch to OFF: system normal startup mode
6. **PSW power button**
   Long press to force power off, short press to soft power off or on
7. **System indicators**
8. **M.2 M key status indicators**
9. **RTC battery holder**
   supports CR / ML 1220 button cell
10. **Dual MIPI interfaces**
    for connecting to DSI displays or CSI cameras
11. **DC power jack**
    DC 5.5 2.1 jack, supports DC 7V ~ 36V power supply
12. **M.2 B key**
    supports connecting 4G/5G communication module
13. **Dual USB 3.2 Gen1 ports**
    USB 3.2 Gen1 Type-A ports, up to 5Gbps
14. **FAN header**
    for connecting cooling fan, 5V, 4PIN JST-SH PWM fan connector
15. **M.2 M key status indicator**
    keeps blinking when NVMe drive is reading / writing
16. **Dual HDMI ports**
    supports dual 4K outputs
17. **PoE header**
    for connecting PoE module, supports PoE power supply through 1000M Ethernet port using PoE switch
18. **1000M Ethernet RJ45 port**
    supports 10M / 100M / 1000M network connections
19. **2500M Ethernet RJ45 port**
    supports 10M / 100M / 1000M / 2500M network connections
20. **USB to 2500M Ethernet chip**
21. **IO-VREF selection**
    CM5 IO logic level: 3.3V (default) or 1.8V
22. **Other function pins**
    Hardware disable function, write protection and chip address selection
23. **TF card slot**
    for connecting a TF card with pre-burnt image (Lite variant ONLY)
24. **NANO SIM card slot**
25. **USB3.0 HUB**



Application Scenarios

Industrial Automation

- Dual Ethernet supports simultaneous PLC + monitoring network
- Wide-range 7–36V DC fits industrial power rails
- GPIO allows direct sensor/actuator integration

Suitable for:

- Production line data acquisition
- Machine status monitoring
- Remote control & maintenance

Smart Edge Computing

With NVMe SSD + optional AI accelerator:

- Handles real-time inference at the edge
- Reduces cloud dependency and latency
- Ideal for intelligent traffic systems, AI vision gateways

Outdoor IoT / Cellular Remote Terminals

- Metal enclosure (BOX-A) protects from dust / moisture
- Active cooling ensures performance in high-temperature environm
- 4G/5G enables deployment where cabling is impossible

Use cases:

- Weather stations
- Remote medical kiosks
- Digital signage / outdoor advertising

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Version Comparison & Purchase Guide

Model	Features	Price (USD)	
CM5-DUAL-ETH-4G/5G-BASE	Board only (no enclosure, no fan)	55.99 (Waveshare) / 59.88 (AliExpress)	
CM5-DUAL-ETH-4G/5G-BOX-A	Metal enclosure + DIN-rail mount + active cooling	69.99 (Waveshare / AliExpress)	

Where to Buy

Waveshare, AliExpress, Amazon

Waveshare, AliExpress

Purchase Considerations

- For prototyping or indoor environments → BASE version
- For 24/7 operation, dusty / hot environments → BOX-A version

Ensure module compatibility before purchase:

- CM5 eMMC version requires USB-C flashing
- CM5 Lite can boot from MicroSD